Fraudulent Claim Detection

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Problem Statement

Every year, thousands of claims are processed by Global Insure, a top insurance provider. But a sizable portion of these assertions are later found to be false, causing large financial losses.
Manual inspections are the company's current method of detecting false claims, and they are ineffective and time-consuming. Fraudulent claims are frequently discovered after the corporation has already made sizable payouts, which is too late in the process. Using data-driven insights, Global Insure hopes to enhance its fraud detection procedure by early in the approval process, classifying claims as either legitimate or fraudulent. In addition to minimizing monetary losses, this would streamline the claims processing procedure as a whole.

Business Objective

- How can we analyse historical claim data to detect patterns that indicate fraudulent claims?
- Which features are most predictive of fraudulent behaviour?
- Can we predict the likelihood of fraud for an incoming claim, based on past data?
- What insights can be drawn from the model that can help in improving the fraud detection process?

Features of the Dataset

- 1000 Datapoints
- 40 features:
- One Target Variable: Fraud_Reported
  - Yes – 247 Count
  - No – 753 Count
- Target variable demonstrates High Class Imbalance. Important to take care of this during modelling.
- Dataset split into Train and Test set for modelling purpose. Train set had 800 datapoints. Test set had 200 datapoints.

Evaluation Metric

- The evaluation metric for this project is precision and recall and ROC Score.
- For business, it is important to identify false positives and true negatives, so recall is to be given more importance.

Data Cleaning and Preprocessing

Changing datatypes and Dropping columns with Unique Values

- Needed to change dtype of certain columns from numerical to categorical as they were categorical features by nature:
  - Number of Vehicles Involved
  - Number of Bodily Injuries
  - Number of Witnesses
- Decided to drop 5 columns as had too many unique values, did not give any useful insights:
  - Policy Number
  - Incident Location
  - Insured Zip
  - Policy Bind Date
  - policy_csl

Imputation of Missing Values

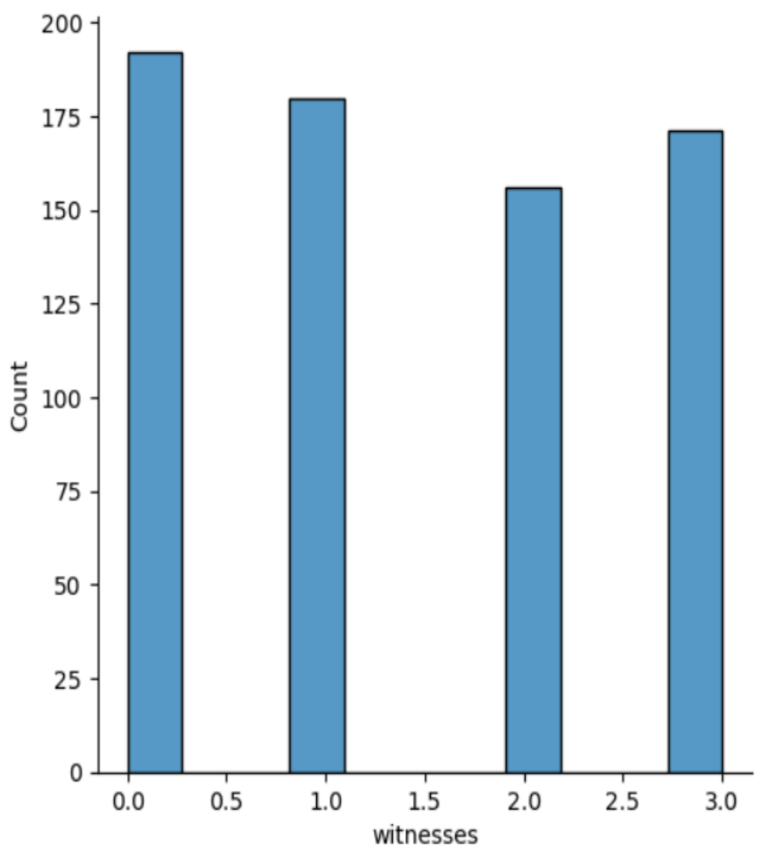
- Three columns had ‘?’ (missing values):
  - Collision Type
  - Property Damage
  - Police Report Available

Label Encoding of Categorical Features

- Categorical features were label encoded before inputting into the model.
- Did not one hot encode because of use of tree- based models.

Exploratory Data Analysis

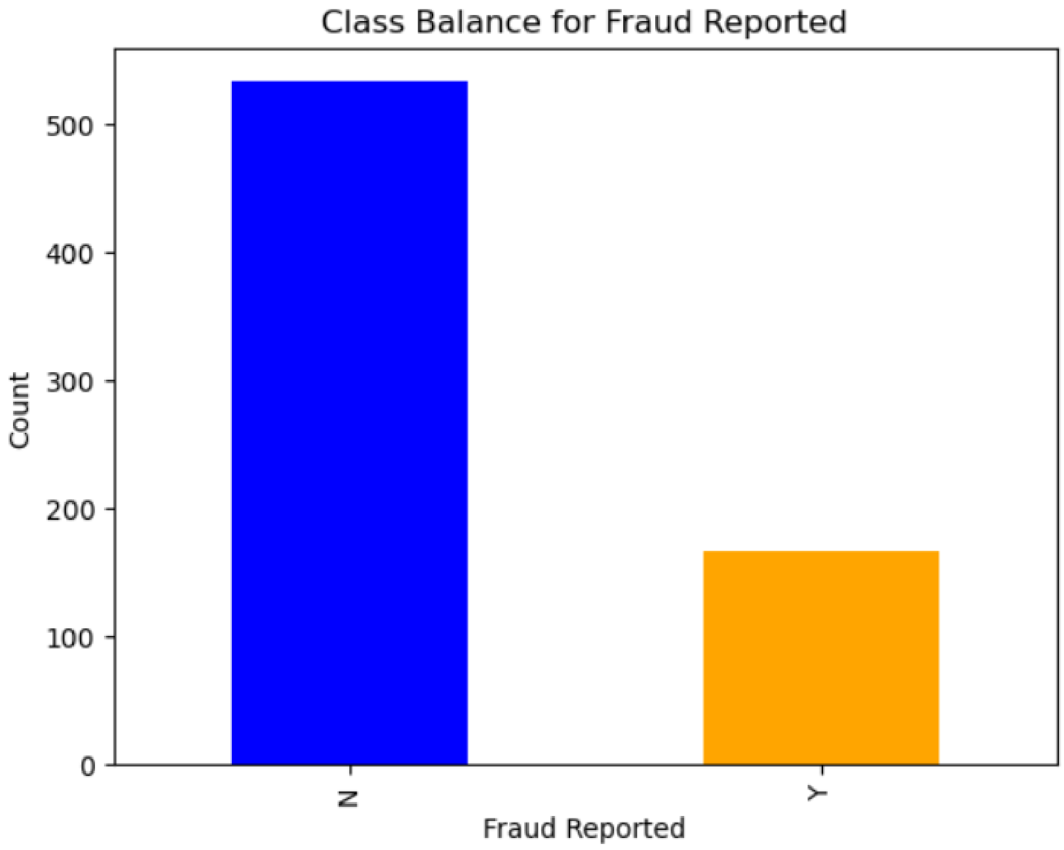
Number Of Witnesses gives some information



Even the presence of one witness increase the chances of detecting a claim as fraudulent.

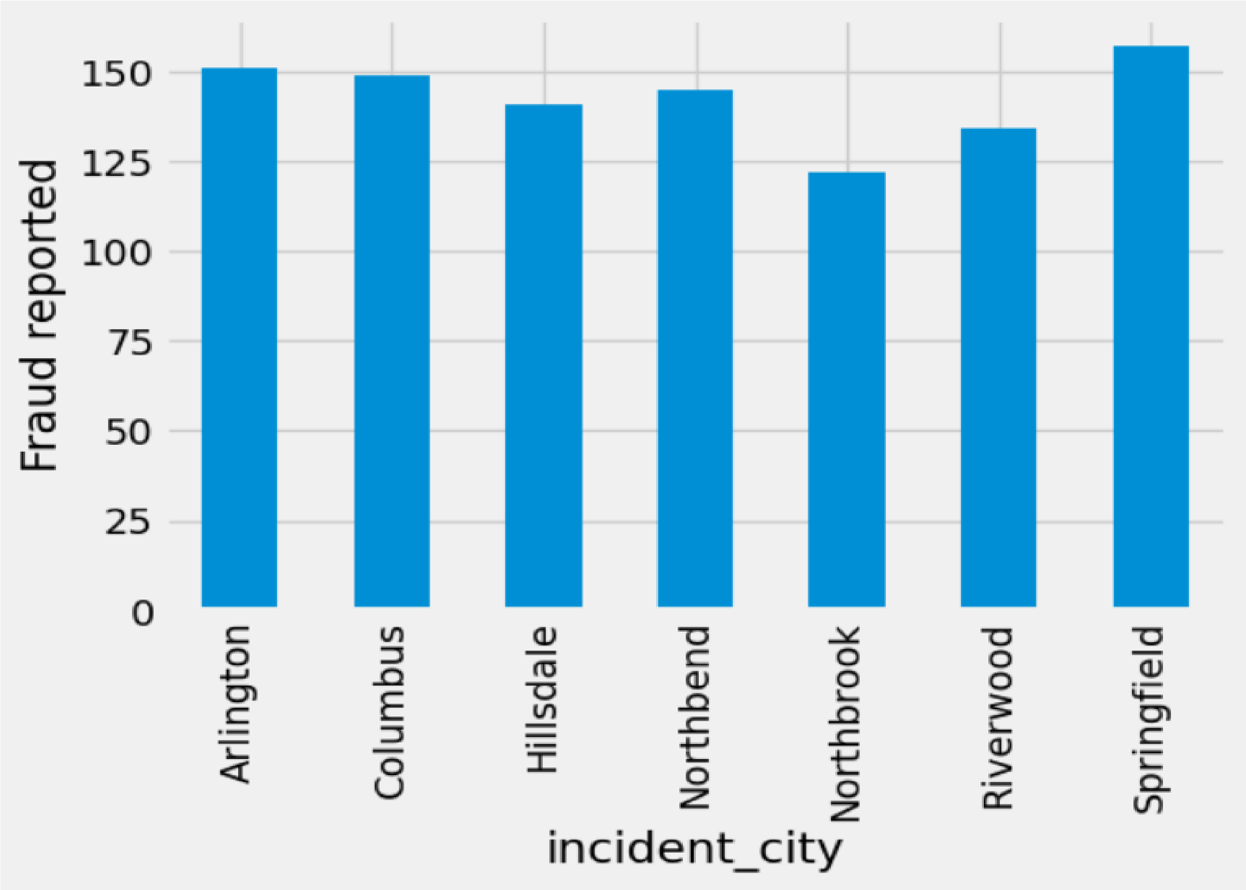
Out of 247 fraudulent claims, 197 claims had atleast one witness present

Fraud Report



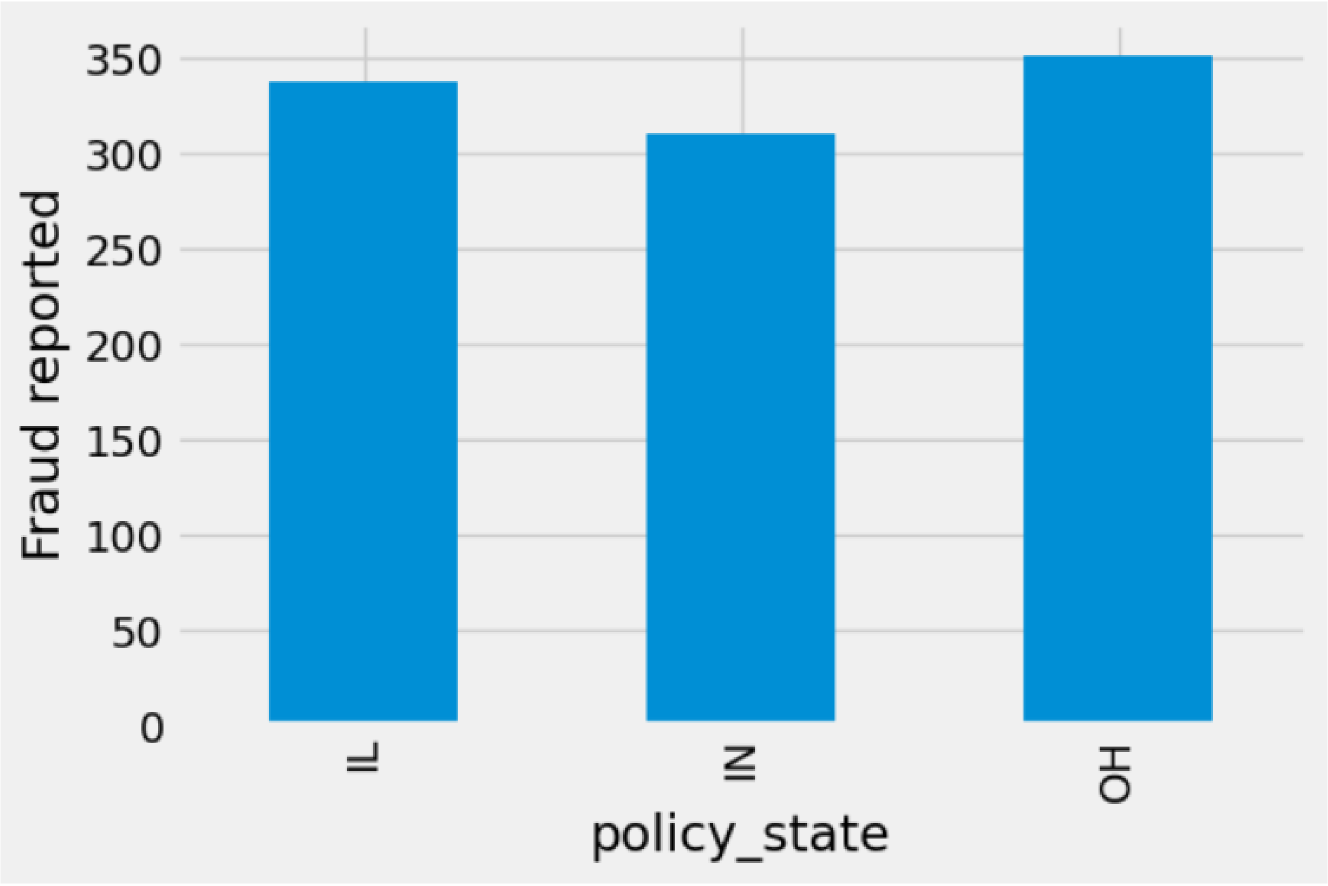
Availability of police report reduces the chances of fraudulent claims, although fewer cases actually have a police report available.
Increase in number of datapoints would help to determine a more clear relationship between fraudulent claims and the availability of police reports.

Fraud Report By City

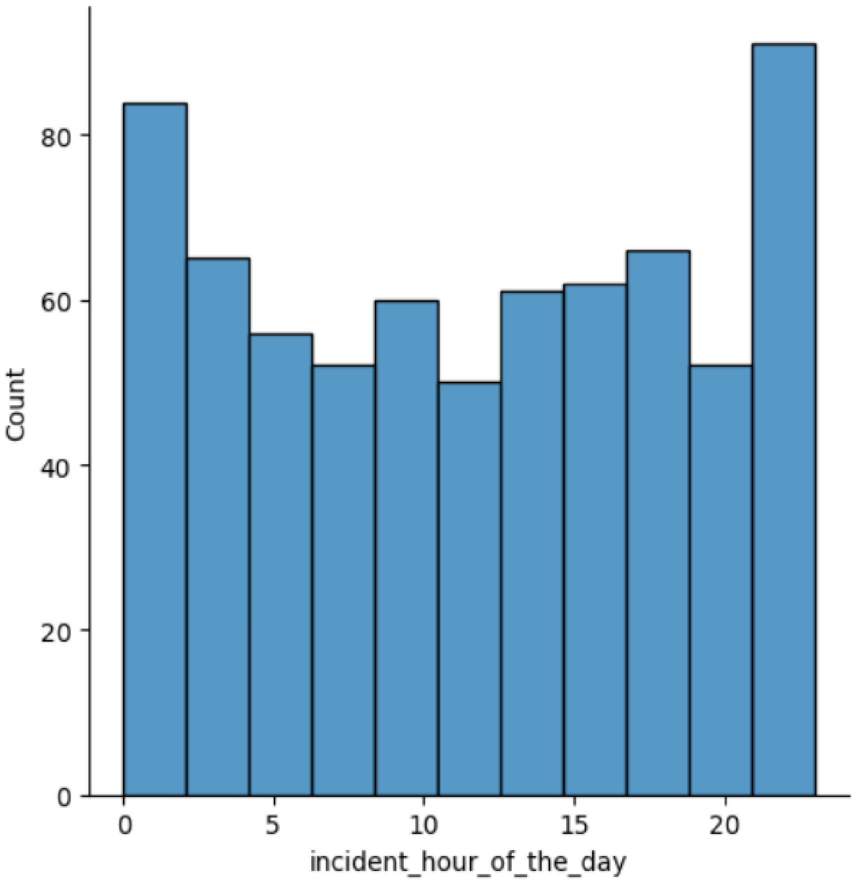


Springfield Contains
155 %

Fraud Report By State



Incident hour of the day

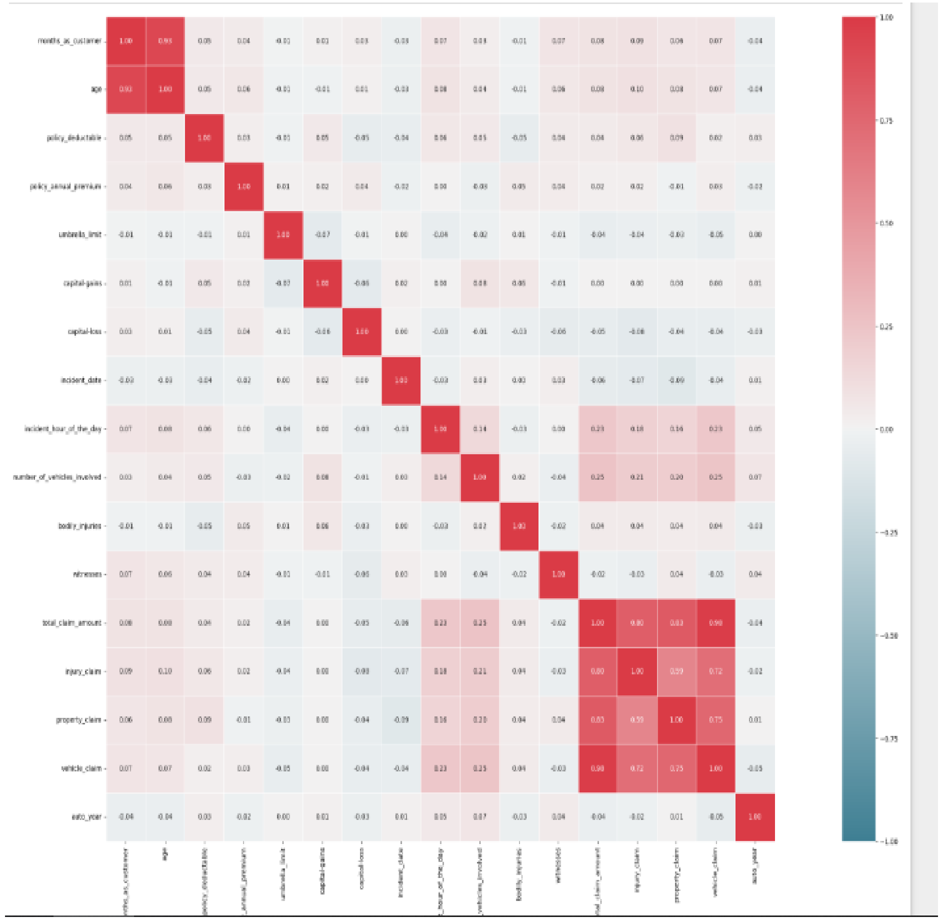


Major Damage is reported fraudulently more often.

This is what would be expected.
People claiming for minor damage tend to add other previous damages that would not actually be part of the incident for which the claim has been made

Feature Selection

Correlation Amongst Numerical Features



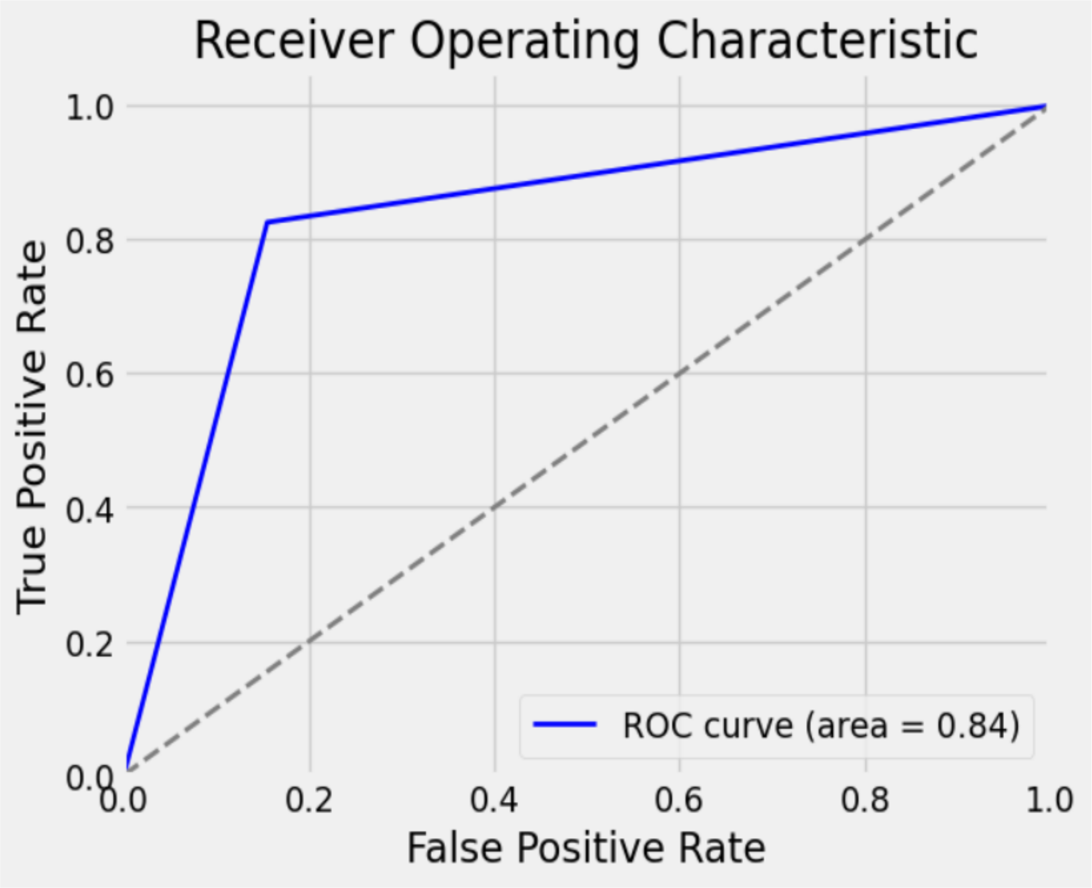
The following heatmap shows high correlation amongst the following columns:
Age and Months as customer Total Claim Amount with its individual constituents. (Injury, Property, Vehicle)
Age and the individual Constituents were dropped from the dataset.

Models and Approaches

Models Used

- After removing correlated features and features with unique values, we have 30 features in the dataset:
  - 9 Numerical Features
  - 21 Categorical Features
- Predictive Model used is RandomForestClassifier and XGBoostClassifier
- Also tried using GradientBoostingClassifier, however above models gave superior performance
- RandomForestClassifier gave best performance, so results for that have been shown.

ROC AUC Curve



ROC AUC Curve shows how well the model has been able to separate out the classes

ROC AUC Score of 0.84 shows the model has done a good job, but there is room for improvement.

Conclusion

- The goal was to detect fraudulent claims,Recall is a good parameter. the summary of logistic regression model and random forest model are below:
- Logistic Regression model:- Training Data Accuracy :- 83.5% sensitivity: 82.5% specificity: 84.5% precision: 84% Recall: 82.5% F1-Score: 83.2%
- (Validation Data) Accuracy :- 83.6% sensitivity: 80.4% specificity: 86.7% precision: 86.4% Recall: 80.4% F1-Score: 83.3% Summary: The fraud is associated with hobbies of chess and cross-fit. It is negatively assosicated with minor damages, total loss. Also, Wrangler owners are good.

- Random Forest model:- (Train Data) Accuracy :- 92.3% sensitivity: 96.3% specificity: 88%precision: 89% Recall: 96% F1-Score: 92%
- (Validation Data) Accuracy :- 82.7% sensitivity: 78.6% specificity: 86.9% precision: 86.1% Recall: 78.6% F1-Score: 82.2% Summary: Again fraud is associated with hobbies of Chess and cross-fit. similar to Logistic regression it also says the claim is negatively associated with minor damage and Total loss. Random forest also asks us to put weight around parameters like vehicle/injury/property claim, a new customer and a loss ratio high.

Insights and Business Decisions

- Claims to be targeted:
  - Having atleast one witness
  - Insured has a hobby of Chess or CrossFit
  - Insureds listing their occupation as Exec- Managers
  - Insureds reporting claims for major damage
  - Claim amount and insured education level are also important features as evidenced by model feature importance
- There is also a need to increase police reports on claims made as it can serve as an initial screen for fraudulent claims
- Also prepared an interactive Dashboard using Dash and plotly showing the Countplots for categorical features

Future Steps

- Time permitting, following additional actions could have been taken:
  - Use of different imputers for missing values to see if any other gives superior performance such as iterative imputer
  - Use of different models like LogisticRegression and SVM to see how dataset responds to these models
  - Identify unique set of features that might always be leading to fraudulent claims.
  - Adding features to dashboard and make it more user-friendly
  - There is a need to capture more data as not many useful business insights can be drawn from 1000 datapoints. More data, better results

GitHub Link